STATEMENT OF WORK (SOW)

Milford, CT Laboratory Salt (Sea) Water Pump and Electrical Infrastructure System Upgrade

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1.0 GENERAL

1.1 BACKGROUND

The Northeast Fisheries Science Center (NEFSC) operates under the National Marine Fisheries Service (NMFS), a part of the National Oceanographic and Atmospheric Administration (NOAA). The agency's core mission is the stewardship of living marine resources.

To support ongoing Milford Laboratory and the Northeast Oyster Breeding Center's facility resilience, process fluid infrastructure, and reliable laboratory operations, the facility requires a complete replacement and modernization of its salt water pumping system and associated electrical infrastructure. The existing system must be decommissioned to make way for corrosion-resistant equipment capable of surviving a harsh coastal environment. The scope of this requirement is directly tied to the success of the Oyster Breeding Center's spawning cycle and research missions. Due to the severe operational impact of any delay, the Government requires strict adherence to the deliverables schedule.

1.2 SCOPE

The scope of this project is the complete decommissioning of existing system, and procurement, installation, and wiring of a new corrosion-resistant salt water pumping system and its supporting electrical distribution and plumbing infrastructure. The Contractor shall furnish all labor, materials, equipment, transportation, and supervision required to execute the following transitions:

- Removal of legacy disconnects, starters, wiring troughs, wiring, breakers, enclosures and plumbing piping.
- Procurement and installation of a Vanton Chem-Gard® non-metallic horizontal centrifugal pump assembly or equivalent.
- Installation and configuration of four (4) active WEG variable frequency drives (VFDs/Inverters) and associated NEMA 4X enclosures or equivalent.
- Upgrading local field instrumentation, current transducers, and electrical distribution wiring.
- Delivery of required on-site backup components (Two spare impellers, one for each pump model) directly to facility inventory.

1.3 OBJECTIVE

To safely remove legacy infrastructure and deliver an operational, highly reliable, and corrosion-resistant salt water pumping system that offers precise local motor control for the Main Building at the Milford Laboratory.

2.0 SPECIFIC REQUIREMENTS / TASKS

2.1 Critical Sequence of Operations (reference section 9.0 Deliverables)

Due to the severe operational impact of current seawater system, the Contractor MUST adhere to the following mandatory sequence of execution:

- **Phase 1 (Procurement):** The immediate order of the WEG VFDs and Vanton Pump assemblies to be delivered within 4 weeks of NTP.
- **Phase 2 (Pre-Demolition Verification):** The Contractor shall **not** initiate any legacy demolition (Task 2) or shut down the existing pumps until all new VFDs, pump assemblies, and critical electrical components are physically on-site and verified by the COR.
- **Phase 3 (Execution):** Upon COR verification of materials, the Contractor may enter the strict 10-day maximum shutdown window to execute demolition, installation, and commissioning

2.2 TASK 1: Legacy Infrastructure Demolition

- **Electrical & Pump Decommissioning:** Safely isolate, apply Lock-Out/Tag-Out (LOTO), disconnect, and completely remove all existing electrical disconnects, motor starters, electrical wiring and plumbing associated with the legacy pump system.
- **Enclosure Removal:** Demolish and remove the existing legacy wire trough, old relay enclosure, and button enclosure.
- **Manual Transfer Switches:** Existing manual transfer switches labeled NORMAL POWER / EMERGENCY POWER are no longer needed and shall be removed. The existing main power panel is on generator power, and no new manual transfer switches are required.
- **Pipe Containment:** Install terminal/junction boxes on the Pumps pipes to securely transition or isolate the existing runs as specified.

2.3 TASK 2: Pump System & Electrical Installation

2.3.1 Pump Operating & Process Fluid Parameters

The procured pump assembly must be rated for continuous duty matching the following parameters for pumps supplying Building #1:

- **Fluid:** Salt Water
- **Specific Gravity (SG):** 1.03
- **Temperature:** Ambient
- **Flow Rate:** 200,000 GPD Gallons Per Day
- **Total Dynamic Head (TDH):** 100 Feet

The procured pump assembly must be rated for continuous duty matching the following parameters for pumps supplying Building #4:

- **Fluid:** Salt Water
- **Specific Gravity (SG):** 1.03
- **Temperature:** Ambient
- **Flow Rate:**350 GPM Gallons Per Minute
- **Total Dynamic Head (TDH):** 10 Feet

2.3.2 Pump Mechanical Specifications

The contractor shall supply and install:

- **Manufacturer/Model:** Two (2) 10HP Vanton Chem-Gard® ANSI horizontal centrifugal pump, Model CGA-PY3X2X6-OP (A-10), brand name or equivalent (To supply building #1).

- **Manufacturer/Model:** Two (2) 15HP **(**Vanton Chem-Gard® ANSI horizontal centrifugal pump, Model CGA-KY4X3X10-OP (A-70) brand name or equivalent (To supply Building #4).
- **Fluid Contact Parts:** Homogeneous,(KYNAR) non-metallic materials.
- **Shaft & Sleeve:** Stainless steel shaft, wet end sleeved with PVDF (Kynar).
- **Materials of Construction:** Kynar (cannot be Polypropylene).
- **Suction & Discharge Connections:** 3" x 2" Flanged.
- **Impeller:** dynamically balanced keyed, semi open design.
- **Shaft & Sleeve:** Stainless steel shaft, wet end sleeved with PVDF (Kynar).
- **Materials of Construction:** Polypropylene (Kynar).
- **Suction & Discharge Connections:** 3" x 2" Flanged.
- **Isolators:** All newly installed sea water pumps shall be placed on anti-vibration pads to minimize vibration on the existing concrete pad. There are no specific specifications on the anti-vibration pads, any anti-vibration pads may be used.
- **Plumbing:** All plumbing piping shall be reconfigured to accommodate the newly installed pumps.
- **Sealing System:** Double Mechanical Seal (#8 & #21).
    - Inner Seal Code: XVLIVLI-1.25-FF
    - Outer Seal Code: XP-ICI-1.25-21

- **Water Jacket:** Flush Face (External sealed water supply required). The contractor is responsible for providing the sealed water supply. The tie-in point is located next to the existing pumps.
- **O-Rings & Gaskets:** Viton.
- **Accessories:** Pedestal and Coupling Guard included.

2.3.3 Electrical Distribution & Wiring Upgrades

- **Breaker Replacement:** Remove legacy breakers and replace with two (2) new GE TED Breakers, brand name or equal, in the main power panel.
- **Feeder Wire Upgrade:** Upgrade and route new properly rated THHN wiring from the power panel to the new disconnect/VFD locations. The existing conduit between the distribution panel and the disconnects in the pump house can be reused.
- **Disconnects:** Provide and install four(4) NEMA 4X non-metallic disconnects (Hubbell, Leviton, or approved equal).
- **Grounding/Bonding:** New NEMA 4X non-metallic enclosures and disconnects can be bonded/grounded at the main power panel located in the same building.
- **Trough Replacement:** Replace the removed legacy trough with a new 8" x 8" NEMA 4X wire trough.
- **VFD Drive Cabling:** Provide, route, and install four (4) properly rated VFD-rated drive cables running from the new inverters directly to the pump motors. Flexible conduit whips, such as seal-tite, are acceptable for connections at motor terminations on anti-vibration pads. .

2.3.4 Motor Drives, Enclosures, & Control Integration

- **Inverter / VFD Installation:** Provide and install four (4) WEG Inverters (VFDs**)**, Two (2)rated at 10HP and two rated at 15HP, NEMA 4X, brand name or equal. NEMA 4 enclosures must be used in the pump house where pumps are located. The adjacent room and shop area cannot be utilized for installing electrical controls and VFDs.
- **Enclosure Upgrades:** Replace the legacy relay and button boxes with a new NEMA 4X Relay Enclosure and a new NEMA 4X Button Enclosure.
- **Pressure Sensors:** Replace four (4) existing Kobold, brand name or equal, pressure sensors and boxes. Provide an additional two (2) spare identical sensor/box units to the COR (6 total sensor units procured).These are purely for local monitoring/displays as stated.
- **Current Transducers:** Provide and install four (4) ACUAMP 3ACT CT units, wired exclusively for motor monitoring only (purely for local monitoring/display as stated).
- **Control Automation:** Wire system relays to manage the Start Control on the WEG inverters. Configure the system for motor speed control to be operated locally via the inverter integrated keypads.

2.3.5 Lighting Systems

- Install one (1) new LED fixture in the designated pump room/area.
- Replace lamps all existing local lighting fixtures in the work zone with energy-efficient LED bulbs.

3.0 CONTRACTOR PERSONNEL

- **Legal Status:** The Contractor shall provide proof that all on-site employees are legal aliens or US citizens.
- **Background Restrictions:** The Contractor shall not employ personnel with felony convictions, or with a record of narcotics use, for work at any NEFSC property.
- **Property Damage Liability:** The Contractor shall be liable for the repair or replacement of any Government property damaged by Contractor personnel.

4.0 OTHER APPLICABLE CONDITIONS

4.1 PLACE OF PERFORMANCE

All work will be performed at the NOAA Northeast Fisheries Science Center (NEFSC) Main Building located at 212 Rogers Ave Milford, CT 06460.

4.2 PERIOD OF PERFORMANCE

- The start date shall be coordinated directly with the Contracting Officer's Representative (COR) and the Contracting Officer (CO).
- The Contractor shall be required to complete all work by February 17, 2027.
- Due to the critical nature of the facility's living marine resources, the Government requires strict adherence to this delivery schedule.

4.3 HOURS OF OPERATION / ACCESS TO THE WORK SITE

- The Federal Government will make best efforts to accommodate working hours outside the set standard 7:00 AM – 4:30 PM (Monday through Friday). The contractor shall observe all federal holidays.
- Any night or weekend work requires prior coordination and explicit approval from the COR. Panel shutdown/outage work may require after-hours or weekend approval, which will be granted if needed.
- All personnel employed within the boundaries of the property or restricted-access areas shall comply with the security regulations for the NEFSC Milford Laboratory.
- The Government reserves the right to close down the job site and order Contractor personnel off the premises in the event of a national emergency or shutdown. The Contractor may only return to the site with verbal approval from the COR.
- The Government reserves the right to exclude or remove from the site any employee of the Contractor or subcontractor whom the Government deems incompetent, careless, insubordinate, or otherwise objectionable.

4.4 EXISTING CONDITIONS

- No as-built electrical drawings, panel schedules, one-line diagrams, control logic diagrams, relay schematics, or hazardous materials surveys are available for the existing system.
- If discrepancies are discovered between the actual existing field conditions and those noted within this SOW , the Contractor shall immediately notify the COR.
- The Contractor shall immediately notify the COR if bidding documents are found to be incomplete, or if discrepancies/omissions are discovered.

4.5 DELIVERY, STORAGE, HANDLING, AND PROTECTION

- Deliver materials to the job site in the manufacturer's original, unopened containers or wrappings with labels, brand names, and installation instructions intact and legible.
- Deliver materials in sufficient quantities to permit continuous, uninterrupted progress.
- Comply strictly with the manufacturer's written instructions for proper material storage. Any damaged materials shall be removed and replaced at the Contractor's sole expense.
- Care shall be taken to ensure total protection of surrounding campus structures, hallways, and interior building elements.

4.6 PUMP ASSEMBLY WARRANTY

- The Contractor shall provide a written comprehensive warranty for service, including labor and materials, for a period of 5 years from the date of system acceptance with no dollar limitation. This must include all associated manufacturer's warranties. **Note:** This 5-year comprehensive warranty applies only to the pump assembly.

4.7 WORKMANSHIP & QUALITY ASSURANCE

- All work shall be of the highest quality, executed by an adequate number of experienced workers regularly skilled in this specific trade.
- Work must strictly align with the manufacturer's published specifications, recent codes, building standards, and the building owner's satisfaction.
- A dedicated Site Superintendent (who must be a direct employee of the General Contractor) must be on-site at all times while work is actively in progress.
- No deviations may be made from this specification or approved shop drawings without prior written approval from the COR. Any installation deviations must be backed by written certification on the manufacturer's letterhead.
- **Substitution Requests:**
    - Substitution requests are not permitted prior to the Notice to Proceed (NTP).
    - Requests must be submitted in writing to the COR post-award. The COR will have seven (7) working days to approve or disapprove.

    - Approved substitutions shall not result in project schedule extensions.
- Contractor shall submit manufacturer cut sheets and literature for all products to the COR for final approval. Quality Control remains the sole responsibility of the Contractor, alongside all incidental repairs.

4.8 FACILITIES AND CONTROLS

- **Waste Management:** Remove and dispose of all debris and trash from the job site in a timely, legally acceptable manner. Perform daily clean-ups to collect all wrappings, empty containers, and paper.
- **Environmental Compliance:** Comply with all applicable Federal, State, and local environmental pollution control and waste disposal laws. Immediately inform the COR of any environmentally hazardous situations.
- **Site Approvals:** Before beginning work, secure COR approval for site access, parking, designated material storage areas, and use of any Government facilities.
- **Campus Rules:** Obey all government requirements for personnel identification and inspection. The NEFSC is a strictly **smoke-free campus**; the Contractor and all employees must adhere to this policy.

4.9 SAFETY & JOB CONDITIONS

- **Safety Plan Submittal:** Five (5) days after NTP and prior to the start of work, the Contractor shall submit a comprehensive Site-Specific Safety Plan (SSSP). This plan must cover open flames, welding, debris management, workplace safety methods, and emergency procedures. It must be accessible on-site at all times.
- **Regulatory Compliance:** The Contractor is solely responsible for safety means and methods and must comply with all current OSHA, NEC, NFPA, EPA, local, state, and federal requirements. Compliance with **29 CFR Part 1910** (OSHA General Industry) and **29 CFR Part 1926** (OSHA Construction Standards) is mandatory.
- **Public Safety:** Ensure work operations do not impact the safety of the public or facility occupants. Implement guards, fences, barricades, flags, and signs at the Contractor's expense wherever hazardous conditions exist.
- **Work Suspension:** If a life-threatening condition is observed by the COR, an oral directive (followed by written confirmation) will be issued to immediately suspend work in the affected area until the condition is corrected.
- **Reporting & Logs:** Submit **Daily CQC Reports** to TPOC Domenick Capomolla summarizing daily progress, weather conditions, and site discrepancies. Strictly observe Lock-Out/Tag-Out (LOTO) regulations.
- **Hazardous Materials & SDS:** Safety Data Sheets (SDS) must be kept on-site. A separate set must be provided to the COR fifteen (15) days prior to starting work. No hazardous material may be brought onto the premises without prior COR consent, and it must be stored in compliance with all environmental laws.

4.10 TECHNICAL CONSTRAINTS

- **System Downtime:** The four (4) existing pumps may be shut down one at a time to maintain seawater flow to the facility for a period of ten (10 Days to complete the project.This shutdown cannot begin until all long-lead items (VFDs and Pumps) are staged on-site.
- **Laboratory Continuity:** Work involving significant vibrations (such as core drilling or crane operations) must be coordinated 48hours in advance to protect sensitive scientific experiments.Either the General Contractor or the electrical contractor may submit this 48-hour advance notice to the COR.

5.0 GOVERNMENT TERMS & DEFINITIONS

- See Appendix 1 & 2 for detailed expansions of applicable professional agencies, standards, and technical acronyms.

6.0 GOVERNMENT FURNISHED RESOURCES

- Reasonable amounts of utility water and electricity for construction purposes will be made available at the designated work site.
- The premises will be occupied by facility staff and scientists during the performance of this work.

7.0 CONTRACTOR FURNISHED PROPERTY

- All equipment, tool/machinery rentals, project materials, manpower, local permits, required testing, and associated fees shall be the exclusive responsibility of the contractor.
- The Contractor shall provide self-contained temporary toilet facilities, coordinating their physical location with the COR.
- The contractor shall provide secure storage for tools and materials. The Government bears no responsibility for providing storage space or security for contractor-owned equipment.

8.0 GOVERNMENT ACCEPTANCE PERIOD

- Inspection of all completed work will be conducted by the NEFSC Facilities Operations Manager and/or the COR.

- Any corrective action identified during government inspection must be successfully resolved prior to final billing processing.
- Deficiency notifications will be issued in writing (or confirmed in writing if delivered orally). The Contractor shall take immediate corrective action.
- The Government will not formally accept the work until it determines that all elements have been satisfactorily completed within the parameters of this contract.
- Acceptance Testing: Required electrical turnover testing includes megger testing, insulation resistance testing, and motor rotation verification.

9.0 DELIVERABLES

The primary deliverable is an operational, modernized salt water pumping station complete with integrated NEMA 4X VFD control controls, upgraded wiring infrastructure, and delivered spare hardware components.

The contractor must maintain open communication with the COR, coordinate the work schedule to ensure continuing facility access, and strictly comply with construction industry best practices. No as-built documentation (updated panel schedules, wiring diagrams) is required at project closeout. The contractor must simply make sure all wiring is correctly labeled.

Deliverables Schedule Table (60 Days Schedule)

Item	SOW Reference	Deliverable / Event	Due By	Distribution
1	2.0,2.1, 4.9	Installation Plan, CSO Schedule, Weekly Updates, All Material and Supplies Documents / Specification Sheets	5 Days after Notice to Proceed (NTP) and prior to start of work.	COR, Contracting Officer
2	4.7	Substitution Requests	Post-Award; COR will have seven (7) working days to approve / disapprove.	COR, Contracting Officer
3	2.1	Materials Delivery Date	Immediate order of the WEG VFDs and Vanton Pump assemblies to ensure	COR

Item	SOW Reference	Deliverable / Event	Due By	Distribution
			delivery within 4 weeks of NTP.	
4	2.1, 4.9	Safety Data Sheets (SDS)	10 Days after Notice to Proceed (NTP) and prior to start of work.	COR, Contracting Officer
5	2.1, 4.9	Site-Specific Safety Plan (SSSP) & Installation/Rigging Plan	Post-NTP; submitted to COR prior to the start of work.	COR
6	2.0	Execution	A strict maximum 10-day shutdown window to execute legacy infrastructure demolition, pump and electrical installation, and commissioning. The four existing pumps may be shut down one at a time to maintain seawater flow.	COR
7	8.0	Final Inspection	Upon completion; walkthrough with COR.	COR
8	4.6	5-Year Pump Assembly Warranty Document	Prior to final project acceptance.	COR, Contracting Officer
9		Post-Award Conference	Day 60	All Parties

APPENDIX 1

PROFESSIONAL AGENCIES, ASSOCIATIONS, CODES, COUNCILS, INSTITUTES, AND STANDARDS

- **ARI:** Air-Condition and Refrigeration Institute
- **AMCA:** Air Movement and Control Association International, Inc.
- **ABMA:** American Bearing Manufacturers Association
- **ACI:** American Concrete Institute
- **ALI:** American Ladder Institute
- **AISC:** American Institute of Steel Construction
- **ANSI:** American National Standards Institute
- **ASCE:** American Society of Civil Engineers
- **ASTM:** American Society for Testing and Materials
- **ASME:** American Society of Mechanical Engineers
- **AWS:** American Welding Society
- **CFR:** Code of Federal Regulations
- **DEP:** Department of Environmental Protection
- **EIA:** Electronic Industries Alliance
- **EPA:** Environmental Protection Agency
- **ISO:** International Organization for Standardization
- **IBC:** International Building Code
- **IEEE:** Institute of Electrical and Electronics Engineers
- **NACE:** National Association of Corrosion Engineers
- **NEC:** National Electrical Code
- **NFPA:** National Fire Protection Association
- **NIST:** National Institute of Standards and Technology
- **OSHA:** Occupational Safety and Health Act / Administration
- **NEMA:** National Electrical Manufacturers Association
- **RCRA:** Resource Conservation and Recovery Act
- **UL:** Underwriters Laboratories
- **USACE:** United States Army Corps of Engineers

APPENDIX 2

ACRONYMS

- **AST:** Aboveground Storage Tank
- **AWG:** American Wire Gauge
- **BMP:** Best Management Practices
- **CO:** Contracting Officer
- **COR:** Contracting Officer's Representative
- **COTR:** Contracting Officer's Technical Representative
- **CT:** Current Transducer
- **DEMARC:** Demarcation Point
- **DEMO:** Demolition
- **DWG:** Drawing
- **FT:** Feet
- **FM:** Factory Mutual
- **GA:** Gauge
- **GC:** General Contractor
- **GPM:** Gallons per Minute
- **HP:** Horsepower
- **IN:** Inch
- **IT:** Information Technology
- **LF:** Linear Feet
- **LOTO:** Lock-Out / Tag-Out
- **MAX:** Maximum
- **NTP:** Notice To Proceed
- **NTS:** Not To Scale
- **OC:** On Center
- **OD:** Outside Diameter
- **PSI:** Pounds per Square Inch
- **PVC:** Polyvinyl Chloride
- **PE:** Professional Engineer
- **PLC:** Programmable Logic Controllers
- **SCH:** Schedule
- **SF:** Square Feet
- **SG:** Specific Gravity
- **SQ:** Square
- **TDH:** Total Dynamic Head
- **THHN:** Thermoplastic High Heat-Resistant Nylon-Coated Wire
- **THK:** Thick
- **TYP:** Typical
- **UST:** Underground Storage Tank
- **VFD:** Variable Frequency Drive
- **VOC:** Volatile Organic Carbon